## Guidelines for Conceptual Design Submission and Judging Criteria

### A. Additional Details of Problem Statement

Confined working envelope < 1.2-meter height (Narrow Body (NB) aircraft Cargo Hold space), Irregular baggage geometry, material and fragility, 20–32 kg, High throughput target ≥ 12 bags/minute, floor load limits ≤ 600 kg/m².

### B. Conceptual Design Documents and the Formats for Submission

1. **Design** diagrams/drawings/3D CAD Model (STEP/STP): Overall concept, major functional elements
2. **Animation** / Video (Max 2 minutes) -MP4: Demonstrate how the proposed system operates.
3. **Report** of design concept (5–8 pages) -PDF: Concept Overview, System Architecture, Mechanism / Key Design Details, Loading and Unloading Operations, Safety, Advantages and Limitations with necessary pictures/diagram etc. Can add more design solutions pictures (different views etc) as Annex if required, at end of the report. Add a cover page with Team name, Lead and Members name and email/contact details, Institute etc.
   - An optional **Presentation** PPT file also can be added with design pictures and annotations.

### C. Judging Criteria

Innovation & Creativity (20%), Functional Effectiveness & Baggage Handling Strategy (20%), Feasibility & Technical Soundness (15%), NB Aircraft Cargo-Hold Compatibility & Space Utilisation (15%), Autonomy & Intelligent Operation (15%), Safety, and Human Interaction (15%)

### D. Design Files Structure for Submission

- Single Zip file -File naming: Team Name_Lead Name(First name as registered), (E.g., Team Name: Robo, Full Name: Sathian Pookkuttath : Robo_Sathian.ZIP)
- Design files/documents naming. File name for each file: Item No_File Category_ Team Name_Lead Name.File type.

  **Robo_Sathian.ZIP**
  - 1_Design_ Robo_Sathian.STP
  - 2_Animation_ Robo_Sathian.MP4
  - 3_Report_ Robo_Sathian.PDF
  - (Optional) 4_Presentaion_ Robo_Sathian.PDF (convert PPT to PDF)
- **Submit your design ZIP file** to *sathian_pookkuttath@sutd.edu.sg*
- Email Subject: Hackathon_Team Name_Lead Name
- All submissions should be made by the Team Lead, who will also serve as the primary point of contact for all communications.

- You may upload your design to any **accessible and shareable repository** and provide/email us with a **valid link** from which we can download the files.

**Submission Deadline: 19 October 2026, Monday**

*For clarifications, please write to sathian_pookkuttath@sutd.edu.sg*

All the very best!

ROAR Lab Hackathon Organising Committee